Readings Guide

Bayesians in the Catallaxy — shared reading for all coauthors

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Re: Bayesians in the Catallaxy: From Mises–Hayek to de Finetti

How to use this guide

This document curates the readings that anchor the paper's argument. It is organized in two parts:

1. A canonical core that every coauthor should read first. These are the four texts on which the paper's central claim — arbitrage trinitarianism and the Mises–Hayek–de Finetti synthesis — most directly rests.
2. Three lane-specific sections mirroring the collaborator briefs in `notes/collaborator-briefs/`. Read your lane after the canonical core. Skim the other two lanes so you know what your collaborators are working with.

Entries follow a consistent template: a full citation, a paragraph on why the reading is on the list, targeted reading guidance (what to read closely, what to skim, what to skip), and questions to keep in mind as you read. Articles, books, and chapters annotated below are available on the shared Box drive; if anything you need is missing there, ask TJB.

References use the citekeys in `biblio.bib` at the repository root; you can cite them in your own writing with `@Citekey`, e.g. `@HayekAER1945`.

Canonical readings (all coauthors)

The canon is six short pieces that, read in sequence, lay out the paper's epistemic and methodological foundation: Hayek's 1937 statement of the knowledge problem, his 1945 statement of the price system as its answer, Buchanan's 1964 methodological program for studying it, Buchanan's 1979 and 1982 statements of the catallactic and causal-genetic conception of order, and Hayek's 2002 mature statement of competition as a discovery procedure. All six are short — the longest is twenty-two pages.

Hayek, "Economics and Knowledge" (Hayek 1937)

Citation. Hayek, F. A. (1937). "Economics and Knowledge." Economica 4(13): 33–54. [Available on the shared Box drive.]

Why. Hayek's first full statement of the knowledge problem and his break with the equilibrium tradition he inherited. The article asks the question that defines the rest of his career: not what equilibrium looks like, but how the coordination of competing plans is brought about empirically. The paper takes this article as the origin of the causal-genetic tradition we inherit; everything in the canon that follows is a development of it.

What to read. All of it; twenty-two pages. Sections II–IV are the load-bearing ones, where Hayek argues that the equilibrium notion is empirically empty unless we have an account of how plans come into coordination through learning.

Questions.

- What does Hayek mean by saying the question of how knowledge is acquired is the central empirical question of social science?
- Where in the article does Hayek explicitly reject the inherited Walrasian equilibrium concept? What does he replace it with?
- How does the 1937 framing anticipate the more pungent 1945 statement? What is new in 1945 that is not already here?

Hayek, "The Use of Knowledge in Society" (Hayek 1945)

Citation. Hayek, F. A. (1945). "The Use of Knowledge in Society." American Economic Review 35(4): 519–530. [Available on the shared Box drive.]

Why. The cleanest twelve pages in twentieth-century economics, and the home text for the paper's "Hayek pillar." The article states the Hayekian epistemic problem — that the relevant knowledge is dispersed, contextual, and not given as data to any single mind — in a form sharp enough that we use it as a touchstone throughout. The paper's claim that exchangeability is the probabilistic formalization of the Hayekian epistemic condition turns on a careful reading of this article.

What to read. All of it; it is twelve pages. Sections III–V are where Hayek argues that the price system is the mechanism by which dispersed knowledge is mobilized; that is the load-bearing part for our purposes.

Questions.

- In what sense is a price a signal rather than a measurement? What is being signaled, and to whom?
- Hayek calls the price system "a kind of machinery for registering change, or a system of telecommunications." How literal is this metaphor? What is the machinery actually doing?
- Where does Hayek's argument implicitly require agents reasoning under genuine uncertainty rather than parametric risk? Mark those passages — they are the ones that resist Arrow–Debreu/Muth–Radner reduction.

Buchanan, "What Should Economists Do?" (Buchanan 1964)

Citation. Buchanan, J. M. (1964). "What Should Economists Do?" Southern Economic Journal 30(3): 213–222. [Available on the shared Box drive.]

Why. The locus classicus of the rejection of "economics-as-engineering." Buchanan argues that the proper object of economic study is not the optimization problem against a known objective function but the catallactic process of exchange. This is the methodological program the paper inherits: Hayek tells us what the price mechanism is doing; Buchanan tells us what economists should be studying. The combination is what makes a catallactic research program intelligible.

What to read. All of it; ten pages.

Questions.

- What is Buchanan's distinction between economy and catallactics? What does it cost to abandon the maximization-against-an-objective framing?
- Buchanan rejects the social welfare function as the proper object of economic study. Is his rejection compatible with reading the catallaxy as a teleological process oriented toward mutual gain through exchange? (We argue yes; the brief for Lane A asks for the philosophical articulation.)
- Where does Buchanan's argument anticipate the move from a single objective probability distribution (the "communism of models") to a population of coherent subjective ones?

Buchanan, "Retrospect and Prospect" (Buchanan 1979)

Citation. Buchanan, J. M. (1979). "Retrospect and Prospect." Concluding chapter of What Should Economists Do? Liberty Fund. [Available on the shared Box drive.]

Why. The short closing essay of Buchanan's 1979 Liberty Fund collection, written fifteen years after "What Should Economists Do?" Buchanan looks back on the catallactic program he proposed and forward to what is still owed. The piece is unusually candid about what the program is for and what kinds of work it asks of economists — read it as Buchanan's own gloss on the canon.

What to read. All of it; under ten pages.

Questions.

- What does Buchanan judge to have been accomplished in the fifteen years since 1964? What does he see as still open?
- Where does the essay sharpen the methodological program in ways that "What Should Economists Do?" left implicit?

Buchanan, "Order Defined in the Process of Its Emergence" (Buchanan 1982)

Citation. Buchanan, J. M. (1982). "Order Defined in the Process of Its Emergence." Literature of Liberty 5(4): 5–18. [Available on the shared Box drive.]

Why. A short, sharp statement of the causal-genetic conception of order: the order is the process of its emergence, not a destination toward which the process tends. This is the philosophical hinge of the paper. The communism-of-models critique, the rejection of Arrow–Debreu/Muth–Radner as a standard, the de Finettian re-orientation — all turn on whether one accepts that the catallactic order is defined in its emergence rather than against an external benchmark. Read this essay slowly.

What to read. All of it; under five pages.

Questions.

- What does Buchanan mean by saying the order is defined in the process of its emergence? What kind of definition is this?
- If the order has no end-state, what is the right philosophical treatment of its telos? (Lane A, question C, depends on this.)
- Where does Buchanan's argument leave room for a quantitative treatment of the catallactic process? Where does it foreclose one? (The paper's answer is that it leaves room for de Finetti but forecloses Walras.)

Hayek, "Competition as a Discovery Procedure" (Hayek 2002)

Citation. Hayek, F. A. (2002). "Competition as a Discovery Procedure." Quarterly Journal of Austrian Economics 5(3): 9–23. [Translation by Marcellus S. Snow of "Der Wettbewerb als Entdeckungsverfahren," 1968. Available on the shared Box drive.]

Why. Hayek's mature statement of competition as a procedure for discovering facts that no one knows in advance — and could not, in principle, know in advance. The paper's claim that price formation is a knowledge-discovery process rather than an equilibrium-revealing one rests directly on this article. Read it as the third panel of the Hayek triptych (1937 frames the problem; 1945 sketches the answer; 2002 says what the answer means methodologically).

What to read. All of it; fifteen pages.

Questions.

- In what precise sense does competition discover facts? Discovers them for whom, and how are they registered?
- Hayek argues that any test of competition's effectiveness must compare it not against an unattainable ideal but against the alternatives we actually have. How does this argument bear on the paper's rejection of Arrow–Debreu/Muth–Radner as a standard against which real markets are judged deficient?
- Where does Hayek's argument require agents reasoning under genuine uncertainty? What would have to be true for the argument to survive a collapse of uncertainty into parametric risk? (Nothing, we think — but the argument needs to be made carefully.)

Strongly Suggested Readings

- See Boettke (1998) for an essential statement about Misesian economic calculation the socialist calculation debate.
- See Horwitz (2004) for a clear statement about what we mean by Misesian microfoundations.
- The Introduction and Chapter 1 of Horwitz (2000) too! (see Horwitz (2020) for a very simple introduction to Austrian economics).
- See Giocoli (2013) for a history of "Savage Bayesian" models in economics.
- It was reading Kiefer and Nyarko (1995) and Bray and Kreps (1987) that made me think of "Bayesians in the catallaxy".
- One very simple way to imagine this project is to read Lachmann (1976) and imagine if he had chosen de Finetti rather than Shackle.

Lane A — Philosophical groundwork (Brittany Oakes & Michael Otteson)

Brief: `notes/collaborator-briefs/brittany-michael-brief.qmd`. Your questions are A (mathematical social ontology of the price mechanism), B (temporal structure of catallactic action), C (causal-genetic structure and telos of exchange), D (the status of formalization).

- Borrill and Tesfatsion (2011) discuss constructive maths in ace models and social science more generally
  - I think the relationship between constructive maths and social ontology will be prove to be a very important link
- Harper (2010) is an important source!
  - He may take on too many arguments that we do not want to touch, but he is unique in pointing out the social ontology of number question in a market setting.
  - The core philosophical issue is the reality of price (a cardinal) number as a socially coordintating process

Lane B — Mathematical foundations (Erin Beckman & Emmy Heywood)

Brief: `notes/collaborator-briefs/erin-emmy-brief.qmd`. The questions are A (formalizing the Misesian-appraisement / de Finettian-prevision bridge), B (exchangeability as the formal expression of the Hayekian epistemic condition), C (endogenous stochasticity and interacting subjective Bayesians), D (exchangeable search and the option/search isomorphism), E (existing-vs-new math accounting).

- I believe Harper (2010) may prove helpful in establishing the mathematical foundations of the de Finetti-meets-Mises problem. We can ignore philosophical battles that he picks that we don't wish to fight ourselves.
- Lad (1996) Perhaps Uncle Frank will still be of service?
- Borrill and Tesfatsion (2011) focus on constructive mathematics. Probably, this is a rich ground for determining the missing mathematical foundations assumed by Misesian economic calculation and the Hayekian price mechanism.
  - This suggests looking to discrete maths and data structures and algorithms from computer science as hints.
- McCall and McCall (2014) is the one textbook on search theory by its originator (at the sequential form).
  - McCall was the first to suggest "exchangeable search"
  - This history, mathematics, and economics in this book is the foundation for Emmy's thesis.
  - Volume II was never written. What are the missing foundations?
- High (1990) is the starting place for Emmy's thesis. The project on "exchangeable search" picks up where High left off with one key insight: search theory and market process theory belong together once de Finetti is embraced.
- Thomsen (1992) is somewhat similar to High (1990) but focuses more on rational expectations rather than search.

Lane C — Computational statistical catallactics (Jake Gunther & Marc Dotson)

Brief: `notes/collaborator-briefs/jake-marc-brief.qmd`. The questions are A (read and summarize the literature), B (reframe ACE as already on the way to CSC), C (Bayesian agent-based models — Marc's anchor), D (computational and signals-processing infrastructure — Jake's anchor), E (strengthen the options-markets proof-of-concept).

On *Brock-Style* Models

The following quote is from Brock and de Lima (1996):

> "We shall devote part of this article to an argument for a style of research in statistical finance where models inspired by direct theoretical arguments are estimated by computer assisted methods such as MSM and where model adequacy (specification testing) is done by bootstrapping financially relevant quantities under the null. That is to say, the quantities that are inputted into the specification tests are themselves motivated by the type of economic and financial behavior one is trying to study. For example, distributions of statistics gleaned off of trading strategies are bootstrapped under the null model being tested in Brock, Lakonishok, LeBaron (1992), and Levich and Thomas (1993)."

A few notes:

- This quote provides a frame for how to think about "agent-based models."
- This is somewhat different from what are known as ace (for "agent-basec computational economic") models in the exisiting literature.
- Often existing ace models as such are not explicitly built upon the underlying statistical decision problem of the agent(s).
- Often agent behavior is rather ad hoc (see the Durlauf critique)
- MSM (method of simulated moments) is an example of indirect inference which in turn is a kind of frequentist ABC without the prior.
  - Is typically understood that ABC comes out of biology and genetic models
  - Indirect inference was developed in econometrics
- Brock et al. (1992) (BLL) is an early powerful example.
- Figlewski (1989) is the prime example and the one we will be building directly upon.
  - Figlewski reaches for Monte Carlo out of convenience. He is not thinking of inventing a new methodology.
  - It is obviously an example of what Brock & de Lima are writing about though.
  - His agent might should be called "a frequentist in a rational expectations equilibrium"
  - We want to imagine the options market-maker as a Misesian entrepreneur and a de Finettian statistical decision maker (i.e. a "Bayesian in the catallaxy")
- See also Sullivan et al. (1999) for an example with the "bootstrap data snooper"

  - They introduce more statistical sophistication relative to BLL
  - Is explicitly decision theoretic (they use the $ln(\cdot)$ loss function or scoring rule
  - Their analysis comes very close to a frequentist version of the evaluation game
  - Their use of the stationary bootstrap avoids strong assumptions of the likelihood function which plague asset pricing studies. It is based upon an argument of exchangeability. I don't believe it is necessary to interpret it as objective/relative frequentist.
  - This comes very close to Geweke (2010), Geweke and Amisano (2011), and Hill (1994).
  - It is an important starting place for the evaluation game study that comes next for us.

- See also Elliott and Timmermann (2016)

On Agent-Based Computational Economic Models

- Delli Gatti et al. (2018) is a practical textbook approach to ace models from a more macroeconomic orientation. Lots to learn from here. And this is the benchmark for standard ace models.
- Tesfatsion (2017) is an important reference for ace models.
  - She ties ace models to extensive form games.
  - She states that ace is a method and not a theory.
  - Pay careful attention to her discussion of stochasticity (we will replace with "endogenous stochasticity")
- Borrill and Tesfatsion (2011) discuss the role of constructive maths in ace model context

The Durlauf Critique

Durlauf (2005) provides an important critique of ace models under the heading of "complexity economics"

The following quote from Durlauf is a stinging critique of the behavioral assumptions of typical ace models:

> "The weaknesses of the behavioural assumptions that underlie complex systems models of economic phenomena are particularly apparent in the context of financial markets models. Many such models, a good example of which is Lux and Marchesi (1999), have been constructed to produce fat-tailed distributions in returns and long memory in volatility, two features whose empirical salience LeBaron (2001) has questioned; a similar exercise is developed in Farmer and Joshi (2002). Regardless of whether these properties are or are not present in financial data, a common problem with these sorts of models is that their logic does not embody what I believe are fundamental features of financial markets. In particular, these models often fail to embody behavioural assumptions that ensure the nonexistence of arbitrage opportunities. In my view, no arbitrage constitutes the most fundamental empirical feature of asset markets and so is properly a building block of the modern theory of finance. The standard theory of finance has little to say about fat tails and long memory in volatility; the theory is not inconsistent with these features but rather does not provide any insight as to why they might occur. A

> successful advance of the current theory of finance should therefore show how to augment a model that reflects no arbitrage in ways that explains these additional features. Instead, behaviour that will eliminate arbitrage opportunities is often not built into these models and presence or absence of arbitrage opportunities is given inadequate attention in the analysis of equilibrium asset returns produced by the model. It is not an advance, in my view, to specify an environment in which arbitrage opportunities go unexploited; it is simply bad microeconomics."

A few notes:

- I agree with Durlauf that ace models that do not model arbitrage represent "bad microeconomics"
- But his critque cuts two ways! Notice how easily he slips from "arbitrage" to "no-arbitrage".
- This is an example of what Buchanan (1964) means about "crossing the bridge" between the individual and the collective.
- Durluaf's critique applies with full force to rational expectations and Nash equilibrium models in market microstruture which have to invent noise traders to generate any trading volume in their models at all.
- Actual arbitrage behavior is mostly missing in financial market models.
- Although see Figlewski (1989) and Figlewski (2017) who clearly understands options market-making as arbitrage. This is a key insight!
- See Fisher (1983) (esp Chapter 4) on why arbitrage requires disequilibrium awareness
- The latter is something financial economists, macroeconomists, and game theorists have not been able to deal with at all
- The result is the no-trade theorems. See also Serrano-Padial (2018).
- This is a real issue that CSC seeks to address head on.

On Austrian Economics and ACE Models

- See Axtell (2017)
- See Boettke and Veetil (2016) (this relates to "endogenous stochasticity")
- See Vriend (2002) for an argument that Hayek would have been an "ace" had such tools been available to him.
- See Devereaux and Wagner (2018) for a similar argument for Buchanan.

A starter reading list — estimation of agent-based models (Marc, question C)

The discussion above is about which models we care about and why; this list gathers the concrete papers on how such models are taken to data — the methods lane behind the MSM / indirect-inference remarks above. The arc to read for is the move from calibration to estimation to Bayesian estimation: exactly the path from ACE toward CSC. All are on the shared Box drive (paired with their `.bib` entries under `readings/ace/`).

Heterogeneous-beliefs / adaptive ABM asset pricing — the structural core: boundedly rational agents switching between competing predictors, producing endogenous fluctuations without exogenous shocks.

- Brock, W. A. (1993). "Pathways to Randomness in the Economy: Emergent Nonlinearity and Chaos in Economics and Finance." Estudios Económicos 8(1): 3–55. (Brock 1993)
- Brock, W. A., & Hommes, C. H. (1998). "Heterogeneous Beliefs and Routes to Chaos in a Simple Asset Pricing Model." Journal of Economic Dynamics and Control 22(8–9): 1235–1274. (Brock and Hommes 1998)
- Brock, W. A. (2000). "Whither Nonlinear?" Journal of Economic Dynamics and Control 24(5–7): 663–678. (Brock 2000)
- Woodford, M. (2000). "An Interview with William A. Brock." Macroeconomic Dynamics 4(1): 108–138. (Woodford 2000) (Orientation to the program, not the technique.)

Discrete choice with social interactions — the bridge from the asset-pricing models to mainstream microeconometrics: the "intensity of choice" logit is a discrete-choice model with interaction effects, which is where Marc's Bayesian discrete-choice toolkit attaches.

- Brock, W. A., & Durlauf, S. N. (2001). "Discrete Choice with Social Interactions." Review of Economic Studies 68(2): 235–260. (Brock and Durlauf 2001a)
- Brock, W. A., & Durlauf, S. N. (2001). "Interactions-Based Models." In Handbook of Econometrics, vol. 5, ch. 54, pp. 3297–3380. Elsevier. (Brock and Durlauf 2001b)
- Anderson, S. P., de Palma, A., & Thisse, J.-F. (1992). Discrete Choice Theory of Product Differentiation. MIT Press. (Anderson et al. 1992)

Simulation-, moment-, and Bayesian-based estimation — read in order: the moment-matching origins, the heuristic-optimization objective-function work, then the explicitly Bayesian treatments (Marc's anchor).

- Duffie, D., & Singleton, K. J. (1993). "Simulated Moments Estimation of Markov Models of Asset Prices." Econometrica 61(4): 929–952. (Duffie and Singleton 1993)
- Lee, B.-S., & Ingram, B. F. (1991). "Simulation Estimation of Time-Series Models." Journal of Econometrics 47(2–3): 197–205. (Lee and Ingram 1991)
- Gilli, M., & Winker, P. (2003). "A Global Optimization Heuristic for Estimating Agent Based Models." Computational Statistics & Data Analysis 42(3): 299–312. (Gilli and Winker 2003)
- Winker, P., Gilli, M., & Jeleskovic, V. (2007). "An Objective Function for Simulation Based Inference on Exchange Rate Data." Journal of Economic Interaction and Coordination 2(2): 125–145. (Winker et al. 2007)
- Franke, R. (2009). "Applying the Method of Simulated Moments to Estimate a Small Agent-Based Asset Pricing Model." Journal of Empirical Finance 16(5): 804–815. (Franke 2009)

- Franke, R., & Westerhoff, F. (2012). "Structural Stochastic Volatility in Asset Pricing Dynamics: Estimation and Model Contest." Journal of Economic Dynamics and Control 36(8): 1193–1211. (Franke and Westerhoff 2012)
- Recchioni, M. C., Tedeschi, G., & Gallegati, M. (2015). "A Calibration Procedure for Analyzing Stock Price Dynamics in an Agent-Based Framework." Journal of Economic Dynamics and Control 60: 1–25. (Recchioni et al. 2015)
- Grazzini, J., Richiardi, M. G., & Tsionas, M. (2017). "Bayesian Estimation of Agent-Based Models." Journal of Economic Dynamics and Control 77: 26–47. (Grazzini et al. 2017) (Marc's anchor — the explicitly Bayesian treatment.)
- Tubbenhauer, T., Fieberg, C., & Poddig, T. (2021). "Multi-Agent-Based VaR Forecasting." Journal of Economic Dynamics and Control 131: 104231. (Tubbenhauer et al. 2021)

Methods references — standing references for the discrete-choice and Bayesian-simulation machinery; consult rather than read cover to cover.

- Cameron, A. C., & Trivedi, P. K. (2005). Microeconometrics: Methods and Applications. Cambridge University Press. (Cameron and Trivedi 2005)
- Train, K. E. (2009). Discrete Choice Methods with Simulation (2nd ed.). Cambridge University Press. (Train 2009)
- Rossi, P. E., Allenby, G. M., & McCulloch, R. (2005). Bayesian Statistics and Marketing. John Wiley & Sons. (Rossi et al. 2005) (Marc's home toolkit for Bayesian discrete choice.)

Logistics

- The bibliography is `biblio.bib` at the repository root. Cite with `@Citekey` — e.g. `@HayekAER1945` renders as Hayek (1945).
- Articles, books, and chapters are available on the shared Box drive. If anything you need is missing there, ask TJB and it will be added.
- This guide is a living document. If you find a reading that should be on the canonical core, or that displaces one already there, raise it.

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Axtell, Robert L. 2017. "Hayek Enriched by Complexity Enriched by Hayek." In Revisiting Hayek's Political Economy, edited by Peter J. Boettke and Daniel J. Sutter, vol. 21. Advances in Austrian Economics. Emerald Group Publishing Limited.

Boettke, Peter J. 1998. "Economic Calculation: The Austrian Contribution to Political Economy." Advances in Austrian Economics 5: 131–58.

Boettke, Peter J., and Vipin P. Veetil. 2016. "Models of Human Action." Cosmos + Taxis 3 (2–3): 45–53.

Borrill, Paul L., and Leigh Tesfatsion. 2011. "Agent-Based Modeling: The Right Mathematics For The Social Sciences?" In The Elgar Companion to Recent Economic Methodology. Edward Elgar Publishing.

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Brock, William, Josef Lakonishok, and Blake LeBaron. 1992. "Simple Technical Trading Rules and the Stochastic Properties of Stock Returns." The Journal of Finance 47 (5): 1731–64.

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Cameron, A. Colin, and Pravin K. Trivedi. 2005. Microeconometrics: Methods and Applications. Cambridge University Press.

Delli Gatti, Domenico, Giorgio Fagiolo, Mauro Gallegati, Matteo Richiardi, and Alberto Russo, eds. 2018. Agent-Based Models in Economics: A Toolkit. Cambridge University Press. https://doi.org/10.1017/9781108227278.

Devereaux, Abigail N., and Richard E. Wagner. 2018. "Emergence, Equilibrium, and Agent-Based Modeling: Updating James Buchanan's Democratic Political Economy." Chap. 6 in James M. Buchanan: A Theorist of Political Economy and Social Philosophy, edited by Richard E. Wagner. Palgrave Macmillan.

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Figlewski, Stephen. 2017. "Derivatives Valuation Based on Arbitrage: The Trade is Crucial." The Journal of Futures Markets 37 (4): 316–27.

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Horwitz, Steven. 2004. "Monetary Calculation and the Unintended Extended Order: The Misesian Microfoundations of the Hayekian Great Society." The Review of Austrian Economics 17 (4): 307–21.

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